White Wine Quality Prediction

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1 Introduction

This project focuses on predicting the quality of white wines based on a range of their features. The primary objective is to develop and present regression models that effectively use various wine characteristics to estimate quality scores. These scores range from 0 to 10 based on sensory data from experts. The significance of this project lies in its potential to enhance understanding of wine quality determinants which could potentially improve production quality and cost. Our approach is constrained by certain limitations: We restrict our analysis to specific models, primarily Linear and Ridge Regression.All computations are conducted exclusively using the NumPy library. This ensures a focused study but also acknowledges the potential for broader analysis with more varied methodologies and tools in future research. The project aims to strike a balance between technical rigor and practical applicability. The aim is providing insights that are both statistically sound and relevant to real-world wine evaluation.

1.1 Project Framework

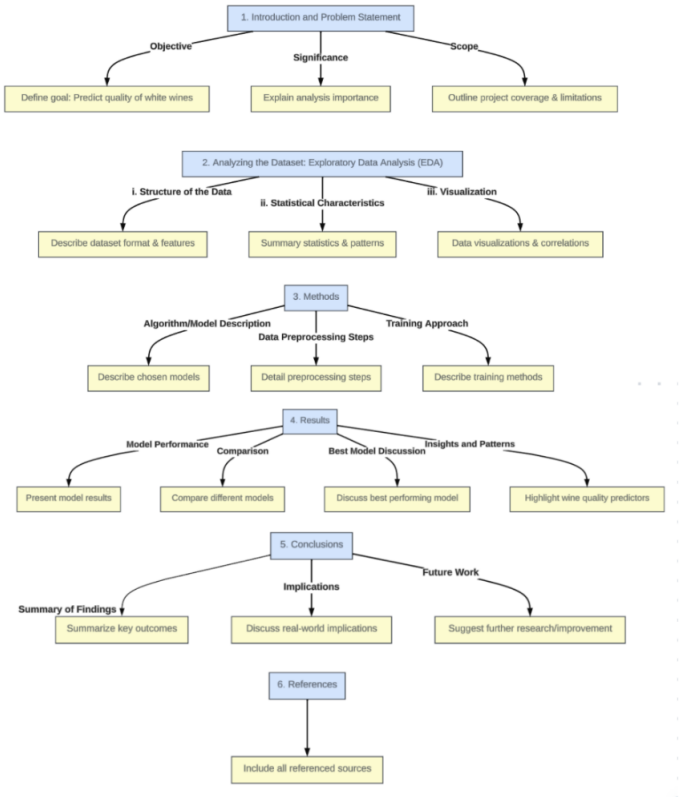


Figure 1: Framework

2 Analysing the Data Set: Exploratory Data Analysis

2.1 Dataset format and features

The first few rows of the dataset (whitewine.head()) display the following columns, representing various 11 physicochemical properties of the wine and its quality score:

Dataset Overview and Integrity: The dataset is well-structured, containing various physicochemical properties of wine, along with a quality score. The absence of missing values implies high data integrity, making it suitable for in-depth analysis without the need for preliminary cleaning steps.

Size and Scope for Analysis: With 4,898 entries and 12 features, the dataset is substantial enough to potentially reveal significant insights into factors affecting wine quality through various statistical and machine learning models.

2.2 Summary statistics and patterns

The statistics of the wine quality dataset provide valuable insights into the features and the target variable (quality) of the white wines. An analysis of the statistical characteristics of features and target variables is given below : **Interpretation and Insights**

Tabulating Feature Statisitics

Physiochemical Data Statistics of White Wine

(I've assumed unitsare same as in Cortez, et al. "Modeling Wine Preferences." Decision Support Systems, vol. 47, no. 4, 2009, pp. 547–553. https://doi.org/10.1016/j.dss.2009.05.016

Attribute (units)	White Wine
Fixed Acidity (g/dm³)	Min: 3.8, Max: 14.2, Mean: 6.9
Volatile Acidity (g/dm³)	Min: 0.08, Max: 1.1, Mean: 0.3
Citric Acid (g/dm³)	Min: 0.0, Max: 1.66, Mean: 0.3
Residual Sugar (g/dm³)	Min: 0.6, Max: 65.8, Mean: 6.4
Chlorides (g/dm³)	Min: 0.009, Max: 0.346, Mean: 0.05
Free Sulfur Dioxide (mg/dm³)	Min: 2, Max: 289, Mean: 35
Total Sulfur Dioxide (mg/dm³)	Min: 9, Max: 440, Mean: 138
Density (g/cm³)	Min: 0.9871, Max: 1.03898, Mean: 0.994
pH	Min: 2.72, Max: 3.82, Mean: 3.1
Sulphates (g/dm³)	Min: 0.22, Max: 1.08, Mean: 0.5
Alcohol (vol.%)	Min: 8.0, Max: 14.2, Mean: 10.4

Quality	
- Min	3
- Max	9
- Mean	around 5.88
- Comment	Most wines are in the middle-quality range, as indicated by the median (50th percentile) value of 6.

Figure 2: Statistical Analysis Overview

The dataset shows a broad range of values for several features like residual sugar, free sulfur dioxide, and total sulfur dioxide, indicating a diverse set of wine samples. The quality scores range from 3 to 9, with a mean score of around 5.88. Most wines are in the middle-quality range, as indicated by the median (50th percentile) value of 6.This analysis lays the groundwork for further exploration, such as examining relationships between features and wine quality.

2.3 Data Visualisation and Correlations

We visualize the distribution of the physio-chemical properties and the quality scores separately. Our objective is to ensure that all visualizations are not only color-blind-friendly but also enhance readability and visual appeal.

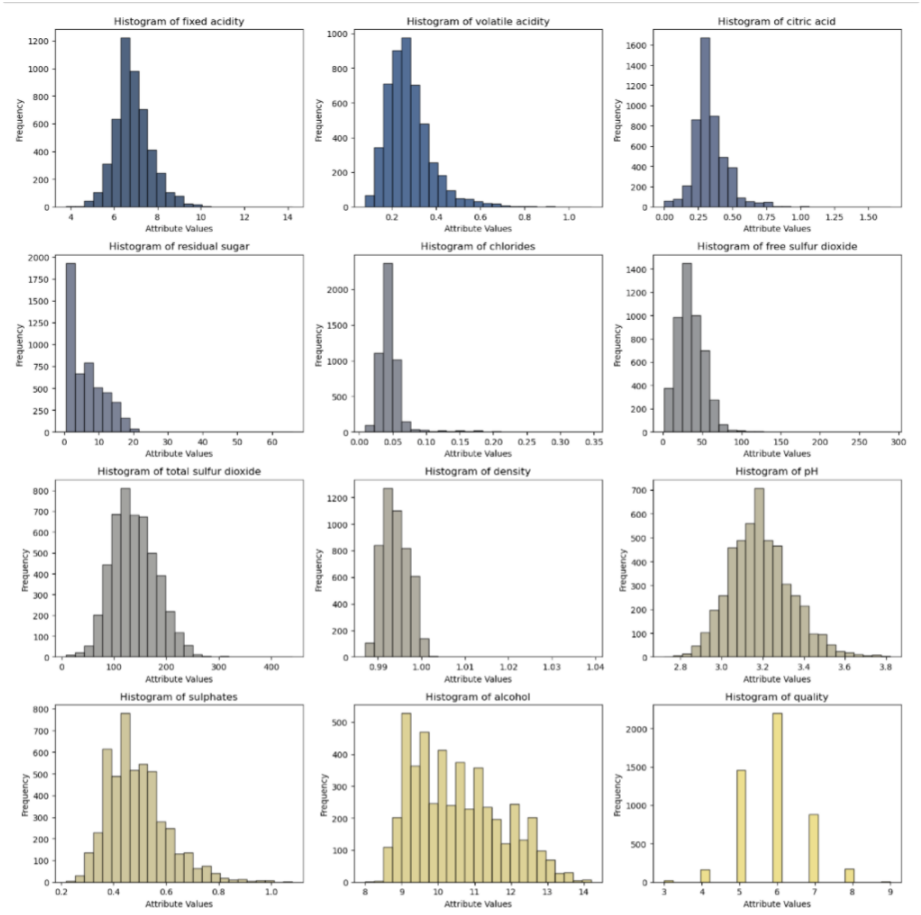


Figure 3: Histograms of each of the physio-chemical properties and quality

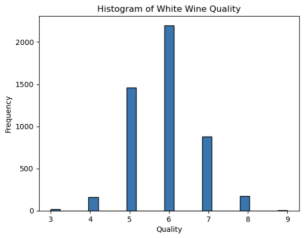


Figure 4: Closer look at Histogram of Quality

Observations

Property	Observation
Fixed Acidity	slightly right-skewed distribution.
Volatile Acidity	Concentrated between 0.2 to 0.4 g/dm³, right-skewed distribution.
Citric Acid	Close to 0.3 g/dm³, right-skewed distribution.
Residual Sugar	Mostly < 10 g/dm³, right-skewed distribution with a long tail towards higher values.
Chlorides	Mostly < 0.1 g/dm³, right-skewed distribution.
Free Sulfur Dioxide	Peak between 20 to 50 mg/dm³, right-skewed distribution.
Total Sulfur Dioxide	100 to 200 mg/dm³, right-skewed distribution.
Density	Normal distribution centered around 0.994-0.996 g/cm³.
pH	Normal distribution centered around 3.2.
Sulphates	Most values concentrated below 0.6 g/dm³, right-skewed distribution.
Alcohol	right-skewed distribution.

Figure 5: Observations on Feature Statistics

Insights

- The physicochemical properties exhibit varied distributions, with many showing a right-skewed pattern, indicating the presence of outliers with high values.
- The quality score histogram indicates that most wines are of average quality (score 5 or 6), with fewer instances of very high or very low quality.

Correlation Analysis and Feature Importance

We continue our analysis by using a heatmap to show how different wine features are related. We also visualise feature importance to find out which ones most affect wine quality, helping us choose the best features for our machine learning models. A notable observation is the strong correlation between free Sulfur Dioxide and total Sulfur Dioxide, as well as between residual sugar and density, signaling a potential case of multicollinearity. This overlap suggests that including both variables in a model might not yield independent information. To avoid multicollinearity issues we consider excluding one of these features.

Interpretations and Insights

- Feature Selection: The correlation analysis highlights Alcohol, Density, Chlorides, and Volatile Acidity as significant predictors. These should be the primary focus in the feature selection for predictive modeling.

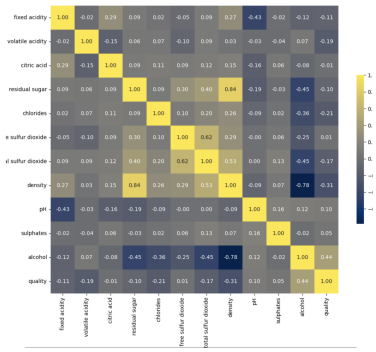
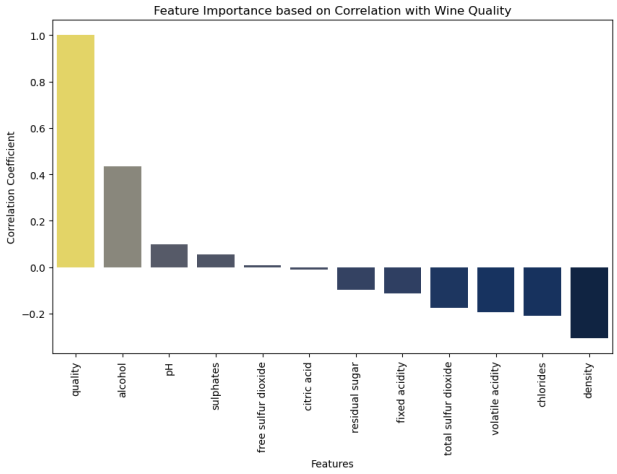


Figure 6: Correlation Heat map



Feature	Correlation with Quality	Insight
Alcohol	Positive	Higher alcohol content correlates with better wine quality.
Density	Negative	Lower density is associated with higher quality wines.
Chlorides	Negative	Higher chloride content may be linked to lower wine quality.
Volatile Acidity	Negative	Lower volatile acidity corresponds to higher quality ratings.
pH	Weak Negative	Slightly more acidic wines tend to have higher quality ratings.
Free Sulfur Dioxide	Weak	Sulfur dioxide levels have a weak impact on quality ratings.
Total Sulfur Dioxide	Weak	Total sulfur dioxide shows a weak correlation with wine quality.
Citric Acid	Very Weak	Citric acid has minimal direct linear relation with quality.
Fixed Acidity	Very Weak	Fixed acidity shows a very weak correlation with quality.
Residual Sugar	Very Weak	Residual sugar has a very weak impact on wine quality.

Figure 7: Observations on Feature Importance

3 Methods

3.1 Data Preprocessing

1. **Loading Data**
2. **Feature Selection**: Based on our analytical insights, we choose features like Alcohol, Density, Chlorides, and Volatile Acidity. We will be mindful to exclude one of the highly correlated sulfur compounds to avoid multicollinearity.
3. **Data Splitting**: The dataset is divided into training (80%) and testing (20%) sets.
4. **Feature Scaling**: We standardize features to ensure equal contribution to the model.

3.2 Linear Regression

Model Description and Theoretical Background Linear Regression is a statistical approach that models the relationship between a dependent variable and independent variables. The equation for a multivariate linear regression is:

$$y = \beta_0 + \beta_1 \times Alcohol + \beta_2 \times Density + \beta_3 \times Chlorides + \beta_4 \times VolatileAcidity + \epsilon \tag{1}$$

Where:

- y is the dependent variable (wine quality).
- $\beta_1, \beta_2, \beta_3, \beta_4$ are the coefficients for each feature. (features: Alcohol, Density, Chlorides, Volatile Acidity).
- β_0 is the y-intercept.
- ϵ is the error term, which accounts for the randomness that the model cannot explain.

Model Training

1. **Augmentation**: The feature matrix is augmented with a column of ones for the intercept term.
2. **Parameter Estimation**: Employing the OLS method, we compute model coefficients.

Model Evaluation

1. **Prediction**: The trained model predicts wine quality on the testing set.
2. **Performance Metrics**: R-squared and Mean Squared Error (MSE) are used to assess the model.

3.3 Ridge Regression

Model Description and Theoretical Background Ridge Regression is an extension of linear regression that includes a regularization parameter, α, to control the magnitude of the coefficients. This approach helps in addressing overfitting and multicollinearity. The equation for Ridge Regression is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n + \epsilon + \alpha \sum_{i=1}^{n} \beta_i^2 \tag{2}$$

Where:

- y is the dependent variable (wine quality).
- $x_1, x_2, \ldots, x_n$ are the independent variables (features: Alcohol, Density, Chlorides, Volatile Acidity).
- β_0 is the y-intercept.
- $\beta_1, \beta_2, \ldots, \beta_n$ are the coefficients for each independent variable.
- ϵ is the error term.

- α is the regularization parameter.

Model Training

- **Augmentation:** A column of ones is added to the feature matrix for the intercept term.
- **Parameter Estimation:** Ridge Regression is employed, and model coefficients are computed.

Model Evaluation

- **Prediction:** The model predicts wine quality on the testing set.
- **Performance Metrics:** The model is evaluated using R-squared and Mean Squared Error (MSE).

4 Results

Model Coefficients

Coefficient	Linear Regression	Ridge Regression
Intercept	5.88004084	5.8714
Alcohol	0.48988789	0.4805
Density	0.10012818	0.1189
Chlorides	-0.02536363	-0.0215
Volatile Acidity	-0.19862543	-0.2141

Performance Metrics

Metric	Linear Regression	Ridge Regression
R-squared	0.19053	0.2439
MSE	0.48674	0.5862

Figure 8: Final Results

Coefficients Comparison:
Both models indicate similar trends in the influence of each feature on wine quality. Alcohol has a positive impact. Chlorides and volatile acidity negatively affect the quality.

The Ridge Regression model shows a slightly higher positive correlation for Density and a stronger negative correlation for Volatile Acidity compared to the Linear Regression model. **Model Fit and Accuracy:**

- The Ridge Regression model has a higher R-squared value (0.2439) than the Linear Regression model (0.19053), indicating that it explains a greater percentage of the variance.
- The Ridge Regression model also has a higher MSE (0.5862), suggesting potentially greater average deviation in predictions from actual values compared to the Linear Regression model (0.48674).

Overall Comparison and Best Model:

- While the Ridge Regression model explains more variance, its higher MSE indicates a trade-off between model complexity and prediction accuracy.
- The Linear Regression model, with a lower MSE, suggests more accurate predictions but explains less variance.

5 Conclusion

This project aimed to predict the quality of white wines based on their features. Our process began with a thorough exploratory data analysis, uncovering key patterns, distributions, and relationships in the dataset. We identified significant predictors and addressed multicollinearity concerns. Implementing Linear and Ridge Regression models was the next step.

The Ridge Regression, with its regularization, provided a broader but less precise understanding of wine quality. In contrast, Linear Regression, though simpler, offered more accurate predictions.

This study underscores the importance of a balanced approach in model selection, focusing on both interpret-ability and prediction accuracy.

Future work, including fine-tuning of the Ridge model and the inclusion of additional features, could enhance our capability to accurately predict wine quality. This would ensure that our models are not only theoretically sound but also practically relevant.

Given the inherent complexity of predicting wine quality, future research should explore more advanced machine learning models. Algorithms like Random Forests, Gradient Boosting Machines, and Neural Networks, known for their ability to handle complex and large datasets, could be particularly effective. Employing these models could lead to a significant enhancement in predictive performance.

6 References

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